

GPS Signal Reception and Spoofing Based on Software-Defined Radio Devices

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Abstract—The Global Positioning System (GPS) has become an essential component of many systems that demand precise localization and timing synchronization. However, the security aspects of GPS raise many concerns, since its signals can be spoofed by stronger faked signals generated from low-cost software-defined radio (SDR) devices. In this study, we propose a SDR-based framework for GPS spoofing attack implementation with dynamic location trajectory. Firstly, we present a SDR-based implementation of a GPS receiver and analyze the reception results. Then, we introduce a design and demonstration of a GPS spoofing transmitter that can lead telephone's detected locations to follow a given specific path. The proposed framework can be a useful instrument for developing new GPS signal reception algorithms and examining the GPS anti-spoof ability of a system.

Index Terms—GPS, GNSS, SDR, Spoofing.

I. INTRODUCTION

Recent advancements in software-defined radio (SDR) technology have created a wealth of opportunities for studying and developing new wireless network protocols. SDR is a wireless communication system where the analog RF components are replaced by software embedded or FPGA systems. Consequently, SDR devices can provide a flexible way to convert base-band signals into radio-frequency signals, which is propagated over the air between transmitters and receivers. Indeed, SDR technology is successfully utilized in many practical applications. For example, the study [1] presented the implementation of an autonomous GSM mobile station based on a SDR device (Nuand BladeRF) and a Raspberry Pi 3. The system can support voice call and SMS messaging within the range of 90 meters. Another study in [2] developed a SDR-based radar platform with multi-band and multi-mode functions. It is shown that the proposed radar platform can detect a drone flying at the distance of about 50 m with the speed of 12 km/h. Finally, the study in [3] provided an open-source SDR-based framework for IEEE802.11p. This prototype can be used for further studies of IEEE 802.11p novel physical layer solutions. Thanks to its simplicity and powerful ability, SDR devices are becoming more and more popular for research and educational purposes.

Among the above-mentioned projects, SDR technology can be used for developing the Global Positioning System (GPS). The GPS has become a crucial part of various systems that demand precise localization and timing synchronization. However, the GPS signal power is relatively low at the

user equipment antenna after traveling a long distance from satellites constellations. Therefore, the real GPS signal can be dominated by a fake signal from a SDR device on the ground. Consequently, a GPS module of a system can be attacked by GPS jamming and spoofing techniques. For example, the study in [4] used a SDR BladeRF X40 device to perform a GPS attack on the DJI Phantom 3 Standard. A security vulnerability test was conducted with different waveform types and various distances. For attacking a mobile phone, the study in [5] demonstrated a GPS spoofing technique by a SDR HackRF device and GPS-SDR-SIM application for signal generation. To prevent unauthorized UAV from intruding airport areas, the study in [6] demonstrated various GPS jamming approaches which can limit the navigation ability of the tested device. However, these studies [4]–[6] used only one-point GPS spoofing technique and perform the attack on a simple application for security evaluation. This motivates us to conduct our experiments with multi-point GPS spoofing technique and to perform the GPS attack on the more complicated application.

In this paper, we introduce a SDR-based framework for GPS spoofing attack implementation with dynamic location trajectory. Firstly, we demonstrate a GPS receiver that consists of the SDR B210 hardware device and open-source GNSS-SDR software. The reception results are analyzed and can be real-time displayed on the Google Earth application. The GPS receiver can be used to improve the GPS signal processing algorithms. Secondly, we propose a SDR-based GPS transmitter that can spoof the complicated Google Maps of a mobile phone and lead the detected locations to follow a given specific path. Moreover, we present our algorithm to generate the GPS signal with dynamic spoofing points and share the implementation code in the following link: <https://link.uit.edu.vn/SDRGPS>.

II. SDR-BASED GPS SIGNAL RECEIVER

A. Experimental Setup

This section implements a GPS signal receiver based on a SDR to capture the radio-frequency signal from GPS satellite constellations. The components of the receiver system are shown in Fig. 1. In our equipment setup, we use a laptop Core i3 running Linux operating system (Ubuntu version 22.04) connected to the RF front-end including USRP via USB cable 3.0, a GPS active antenna and a supply power bias-tee 5V DC.

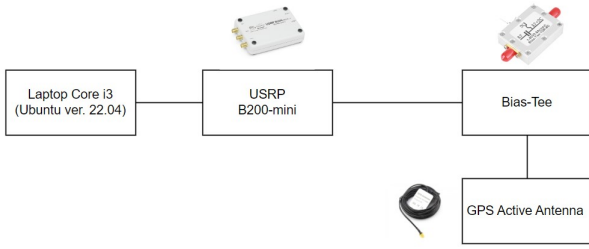


Fig. 1: GPS Signal Receiver Blocks

An open source GNSS-SDR application is installed to capture the signals, which helps with the satellite acquisition, tracking, and navigation solution computation on the host device [8]. The UHD package is required to check the USRP's connection status. This package displays the specifications and collects the information of the SDR device which is connected to the host.

B. Demonstration of GPS Signal Reception

We execute the configuration file using GNSS-SDR software in the terminal, in our demonstration, the following parameters and values are used to collect the signal using L1 band: frequency is 1.5754 MHz, the gain is 40 dB, the sampling frequency is 4 MHz and the input sample rate to the processing channel is 4 MHz. Once the program is executed, the terminal shows up the tracking log of the GNSS-SDR about the current time, the tracking GPS message, the navigation message and the result about the real-time position. The result information consists of longitude, latitude, height, velocity and ENU (East, North, Up) coordinates. In this study, our location to receive the signals is on the 12th floor, E building, University of Information Technology, Ho Chi Minh City, Vietnam. When the receiver successfully detects GPS and decodes a portion of the GPS navigation message frames, the terminal displays the following message:

Position at 2022-Jun-11 07:20:05.500000 UTC using 4 observations is Lat = 10.869766362 [deg], Long = 106.802607988 [deg], Height = 59.903 [m] Velocity: East: 0.92 [m/s], North: 0.004 [m/s], Up = 0.313 [m/s].

After executing the GNSS-SDR software, the data is recorded into KML (Keyhole Markup Language), NMEA (National Marine Electronics Association) and RINEX (Receiver Independent Exchange Format) files, we utilize the KML file and import it to Google Earth to display the location points from the data result, which are shown by Figure 2.

According to the outcome, the dilution of precision (DOP) values which are horizontal DOP is around 1.9, vertical DOP is around 4.2, and position DOP is around 4.6. Based on those values, latitude, longitude, and altitude values may be considered to be utilized to make measurements that are sufficiently precise.

For the purpose of evaluating the receiver's performance, 30 coordinate samples were taken, which the receiver coordinate is represented by x_{ref} and the signal coordinate i_{th} by x_{th} .

As Figure 3, we denote A_i as the distance between receiver point and received point with coordinate i_{th} , B_i and B'_i as the distance between received point and receiver point which are reflected on the earth surface, C_i as the height from receiver



Fig. 2: Received Location Points displayed by Google Earth

point to the earth surface, C_{1i} as the height from coordinate i_{th} point to the earth surface and C_{2i} as the height from received point to receiver point. B_i is calculated by Haversine formula (1), (2), (3), using lat and long variables as the latitude and longitude of two coordinates and R in the formula is Earth's radius = 6371km [9]:

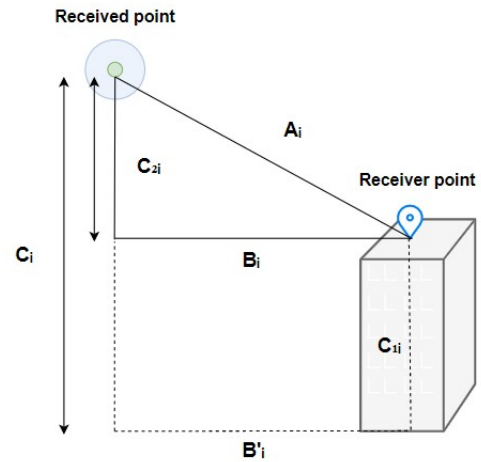


Fig. 3: Distance between receiver point and coordinate received point

$$\Delta_{lat} = lat_2 - lat_1 \quad (1)$$

$$\Delta_{long} = long_2 - long_1 \quad (2)$$

$$B_i = 2R \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta_{lat}}{2}\right) + \cos(lat_2) \cdot \cos(lat_1) \cdot \sin^2\left(\frac{\Delta_{long}}{2}\right)}\right) \quad (3)$$

Since the distance between the receiver and the signal points received is small, we ignore the curve surface from the Haversine and interpret B_i as the line on a flat surface. In Figure 1, A is calculated by Pythagoras with the values of B calculated by Haversine formula and C_{2i} calculated by the

absolute of distance between two heights of signal point and receiver point as formula (4):

$$C_{2i} = |C_i - C_{1i}| \quad (4)$$

After having distance from each receiving point to the receiver point, we calculate the mean square error (MSE) with N is the received signal number by formula (5):

$$MSE = \frac{1}{N} \sum_{i=1}^N D_i^2 \quad (5)$$

The MSE result is 24.17m. To enhance the result, it is recommended that the DOP values must increase their rating by doing the experiment under a clear sky with fewer objects around the receiver's location.

III. SDR-BASED GPS SIGNAL TRANSMITTER

After receiving GPS signals from real satellites and processing them using USRP B200mini, in this section, we propose a solution to help spoof GPS signals using LimeSDR Mini and LimeGPS.

The LimeSDR Mini board is a highly cost-optimized hardware platform for high-performance digital and RF designs. It is a smaller, less expensive version of the original LimeSDR. The LimeGPS is a real-time GPS signal simulator for LimeSDR based on GPS-SDR-SIM. [7].

A. Experimental Setup

In this section, our team builds a system to spoof GPS signals, specifically we use a computer running Linux operating system particular is Ubuntu version 22.04 connected with one board to generate GPS signal LimeSDR Mini on RF frequency between 10MHz to 3.5GHz and RF Bandwidth is 40MHz via USB communication by directly plugging the board device to the computer.

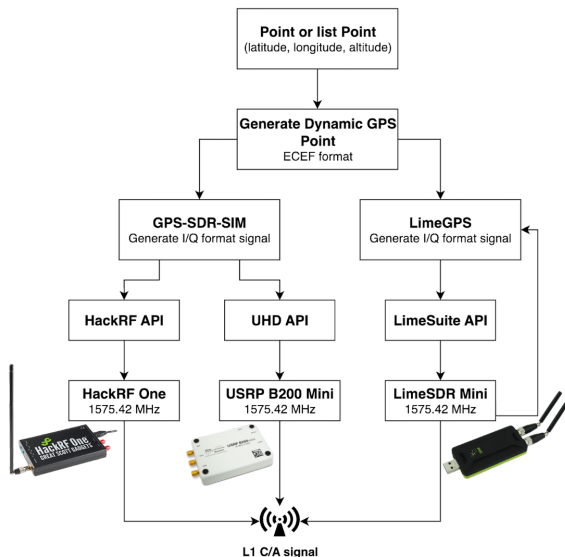


Fig. 4: Processing model for GPS spoofing attacks.

To perform GPS signal spoofing, it is important to assemble a point or group of points with exact coordinates, including latitude and longitude. From the previously collected location data, our team self-developed Python code to generate GPS

signals based on the LimeGPS, which will be covered later in the next section. The LimeGPS application generates signals in I/Q format using points generated by Python code. The LimeSDR Mini then broadcasts the signal using the LimeSuite API library developed by Myriad-RF. Lime Suite is a collection of software supporting several hardware platforms based on the LMS7002M transceiver RFIC, such as the LimeSDR family. The implementation technique is described in more depth in Figure 4.

In addition, our team tested on the USRP B200 Mini and HackRF One. Still, these two devices will be slightly different from LimeSDR Mini in that they both use gps-sdr-sim to generate GPS signals but do not create in real-time and have to enter each stage of the desired manually. To convey the newly generated signal, two distinct boards will each utilize a different API, namely the HackRF One board will use the HackRF API and the USRP B200 Mini will use the UHD API.

B. Generate Dynamic GPS Point

LimeGPS is an open-source software that allows the generation of GPS signals at a specified location based on GPS-SDR-SIM. The 1-bit I/Q-formatted signal will be created, stored in RAM, and then broadcast by LimeSDR Mini using the LimeSuite API. The majority of recently published research concentrate on spoofing the GPS signal in a static location, meaning the user is stationary and not moving [5], [6], [10]. When GPS anti-spoofing techniques are developed and used, it gets harder to trick GPS receivers, therefore transmitting a GPS signal with a fixed location would not qualify applications like Google Maps to trust that signal. After research and experimentation, we have found a method to make apps like Google Maps believe in fake GPS signals emitted by us without disabling the Google Location Accuracy services.

Let's say we need to transmit a spoofed signal of a location where x is latitude, y is longitude, and z is altitude. When we rotate the vector R around the axes OX , OY , and OZ , we designate point I with coordinates (x, y, z) as the centers of the three circles. We then determine the coordinates (x', y', z') of the points I' evenly spaced on each circle.

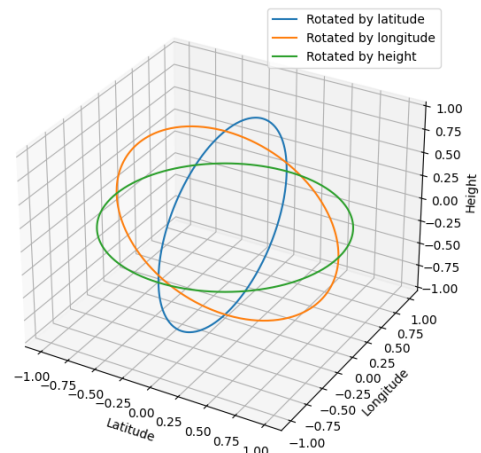


Fig. 5: Unit circle.

The unit circles in figure 5 have a radius R of 11.12 cm. We get 360 points I' on each circle by following formulas (6), (7) and (8) as follows:

$$I'_{OX} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) & 0 \\ 0 & \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y + R \\ z \\ 1 \end{bmatrix} \quad (6)$$

$$I'_{OY} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z + R \\ 1 \end{bmatrix} \quad (7)$$

$$I'_{OZ} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 & 0 \\ \sin(\theta) & \cos(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x + R \\ y \\ z \\ 1 \end{bmatrix} \quad (8)$$

Following the establishment of 1080 points using the aforementioned three formulas, we create a file to store the dynamic coordinates of the fake location in the ECEF format, where each point represents a time slot with a duration of 0.1 seconds. We will make a list of coordinates based on the motion trajectory for the simulated scenario where the user moves from one spot to another, then generate dynamic coordinates for each point in the list. All dynamic coordinates will be saved into a CSV file with the data format of ECEF for use in the generation and transmission of GPS L1 C/A spoofed I/Q signals.

C. Demonstration of GPS Spoofing

The results of testing and evaluating the effectiveness show that the fake GPS signal generator solution proposed by us above gives better results than the existing solutions. We were able to do that because we inherited the strong advantages of LimeGPS and further developed the dynamic signal generation feature based on the unit circles. The evaluation is conducted in the following two contexts: The first is that we have taken on dynamic GPS signal spoofing to force nearby receivers to produce the desired position. Secondly, in order to simulate the user traveling along a specific route from one spot to another without they actually doing so, we also transmit a phony dynamic GPS signal.

The evaluation results of the first context are shown in Figure 6. Since the real location is VNUHCM - University of Information Technology in Ho Chi Minh City, Vietnam, we broadcast a false GPS signal to the nearby phones to trick it into thinking we are at the place we specified in Ukraine. We then tried to find our way back to Vietnam using Google Maps and got the results shown in Figure 6.

In the second context, we simulated a user moving from VNUHCM University of Information Technology to one kilometer-away Ho Chi Minh City Community Center. We use the LimeSDR Mini board to transmit GPS L1 C/A frames after generating dynamic fake GPS signals with the aid of this research's proposed approach. Using the USRP B200mini, we get GPS L1 C/A frames, which then express the locations in the order they were received in Figure 7.

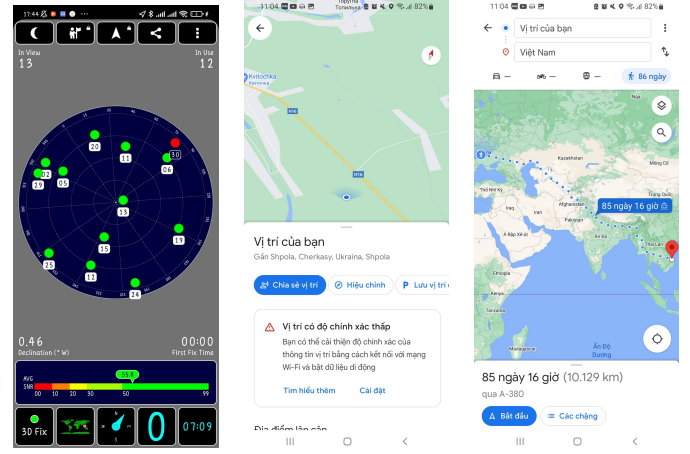


Fig. 6: Attacks spoof the location of smartphones.



Fig. 7: user moving from VNUHCM University of Information Technology to one kilometer-away Ho Chi Minh City Community Center.

When we carried out the GPS spoofing attack in the second scenario, we also videotaped the position shift in accordance with the movement on Google Maps. Video demonstration can be accessed in the following link: <https://link.uit.edu.vn/SDRGPSVideo>.

The smartphones used by us to evaluate the effectiveness of the proposed solution in this research are Samsung Galaxy Note 8 (Android 8.0), Samsung Galaxy A31 (Android 12), Xiaomi Redmi Note 8 Pro (Android 11), Xiaomi Redmi Note 11 (Android 11), Samsung Galaxy S7 Edge (Android 7.0), iPhone SE 2022 (iOS 15). All of these smartphones use 4G, WiFi, and do not disable Google Location Accuracy services.

In the case of reception signals from other GNSS satellites, we can change the configuration file of the GNSS-SDR and then perform the calculation of the current position based on the received data. In the case of a GNSS signal spoofing attack, our team will research and evaluate techniques for generating fake signals for GNSS systems like GLONASS, Galileo, and BeiDou in the future. Utilizing the closed-source, for-pay SatGen software offered by LabSat is now one of the better methods. SatGen software allows us to generate a GNSS RF I/Q format or IF data file based on a user-generated trajectory file that can be replayed on a LabSat GNSS Simulator. This allows us to simulate almost any kind of test, at a set time and date, anywhere in the world. Any SDR that supports the GNSS system frequency range can transmit the GNSS RF I/Q format file.

IV. CONCLUSIONS

In this paper, we proposed a multi-point GPS spoofing attack implementation based on a SDR device and successfully demonstrated the technique on the Google Maps application. The detected locations of a mobile phone are controlled by our framework and guided to follow a given specific path. This solution can be used for analyzing the spoofing vulnerability of a system using GPS. Moreover, we demonstrated a SDR-based GPS receiver implementation and analyzed the reception results. This GPS receiver can be combined with the proposed GPS transmitter to form an effective framework for improving and addressing security issues of the GPS.

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